

THE DRAWBACKS OF NATIVE APPLICATION DEVELOPMENT.

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WHAT IS NATIVE APPLICATION DEVELOPMENT ?



source: google images

One of the most popular ways to create applications, programs, or just software is to make them natively. Native app development means creating a mobile application that is tailored and dedicated to a specified platform like iOS or Android

Native app development means creating a mobile application that is tailored and dedicated to a specified platform like iOS, or Android.

Because native applications are built specifically for the operating system, they provide higher user engagement than hybrid apps. Native mobile apps generally perform and look better than their web-based counterparts, which must serve numerous platforms. Furthermore, native mobile applications have access to device hardware and capabilities, such as sensors and cameras, that are not available via a mobile browser interface alone.

Native mobile apps, unlike websites and web applications, do not operate in the browser. They must be downloaded from platform-specific app stores like Apple's App Store and Google Play. Each program may be accessed by tapping its icon on your device's screen after installation.

Native applications are more complicated to create than mobile websites. There's no need to be concerned about browser compatibility or behavior. You may use the native capabilities of mobile operating systems to create a richer user experience and implement app features.

In other words, native applications are software explicitly created for a single platform. Programming languages and tools designed for a single platform are used to create the

application. You can build an Android app in Java or kotlin while creating an iOS app in Swift and Objective-C.

THE DRAWBACKS OF NATIVE APPLICATION DEVELOPMENT :

- LENGTHY DOWNLOADING PROCESS.
- LESS FLEXIBILITY.
- EXPENSIVE DEVELOPMENT.
- TIME CONSUMING DEVELOPMENT.
- MORE SUPPORT AND MAINTENANCE.
- NATIVE APPS REQUIRE FREQUENT UPGRADES.
- NATIVE APPS SHINE FOR MANY COMPANIES,NOT ALL.

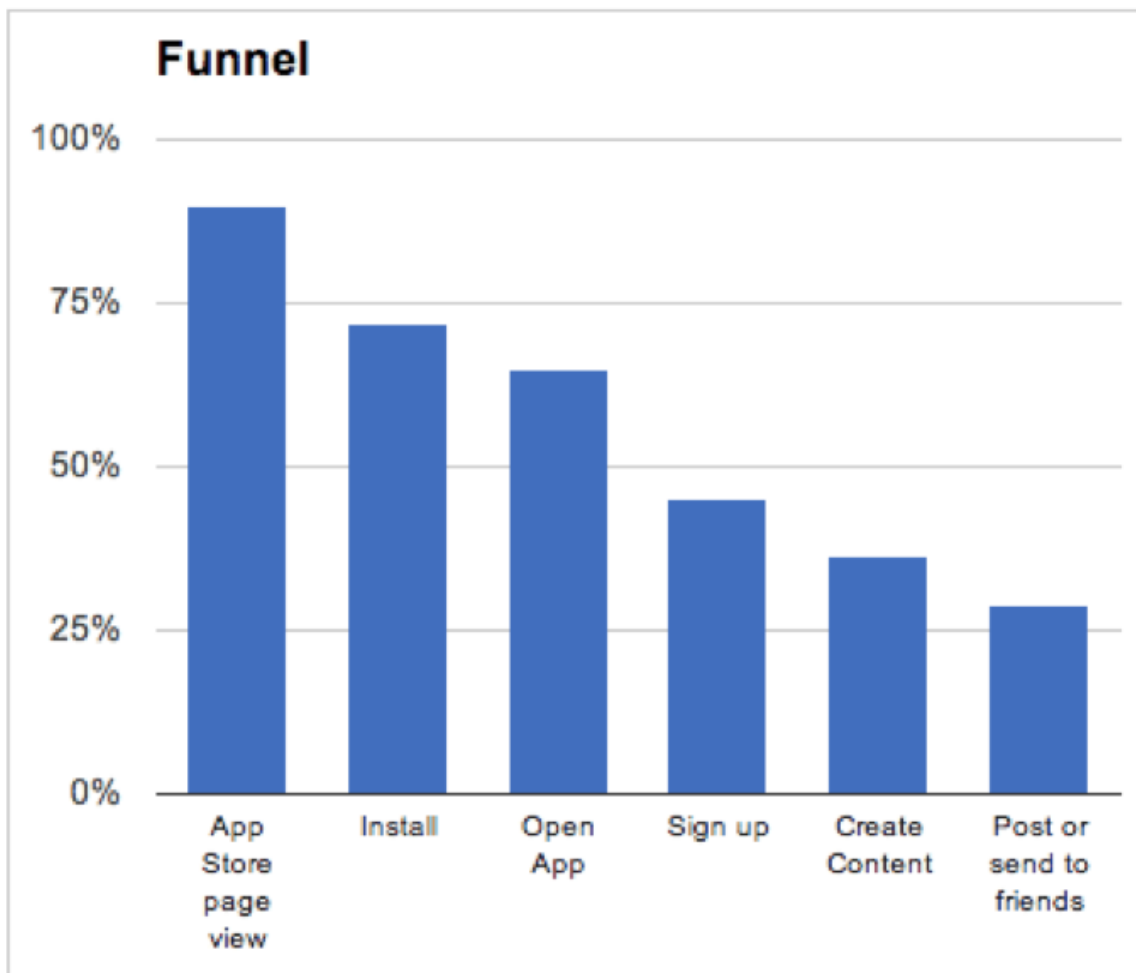
LENGTHY DOWNLOADING PROCESS :

Native apps are ready to use only when users download them from the app store (Google Play or Apple App Store).

This involves a lot of steps – going to the app store, finding the application, accepting its terms and conditions, and then finally downloading it.

Many people don't have the time and patience to take all the steps to download an app. You lose 20% of users at each stage in an app funnel:

- App store pageview
- Install
- Open app
- Sign up
- Create content
- Post or send to friends



In each step of the six-step app funnel, about 20% of users are lost.

LESS FLEXIBILITY :

Developers have very less flexibility regarding the platform for developing native apps.

Developers have to code for one platform at a time, with separate coding for Android and IOS.

Much of the time, if you are hiring app developers for your native app idea, you'll have to hire two teams of developers one for Android and one for IOS.

The best feature of native apps (running native compiled code on top of the OS) is also its biggest weakness.

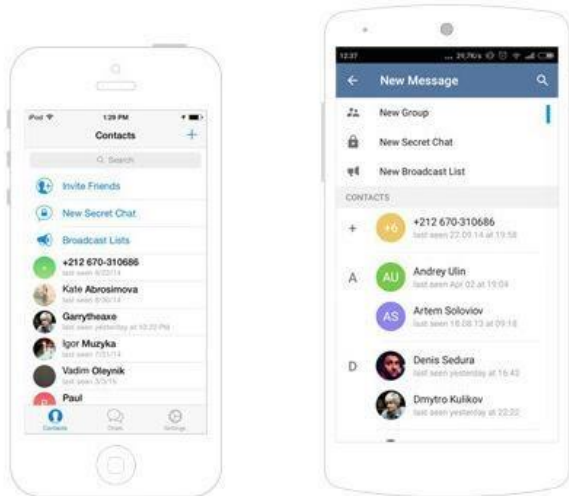
That's because their source code can only be used on one platform and can't be repurposed for another.

Sadly, it's not a matter of copying code and porting it into another platform. You often need to rewrite and tailor your app's logic to the unique nuances and SDK of your target OS.

Moreover, doing this manually is guaranteed to introduce errors.

But more than making the app itself work, the challenge lies in redesigning the UI to fit with the target OS. It needs to fit in with the look and feel of the OS while retaining the user-friendliness and UX of the original.

Achieving this can take a lot of planning and effort, as what Telegram obviously put into the successful Android port of its iOS app:



Sometimes, it's easier to just retain the look and feel of the original app. This is what Instagram did when it retained the look and feel of their iOS app to Android.

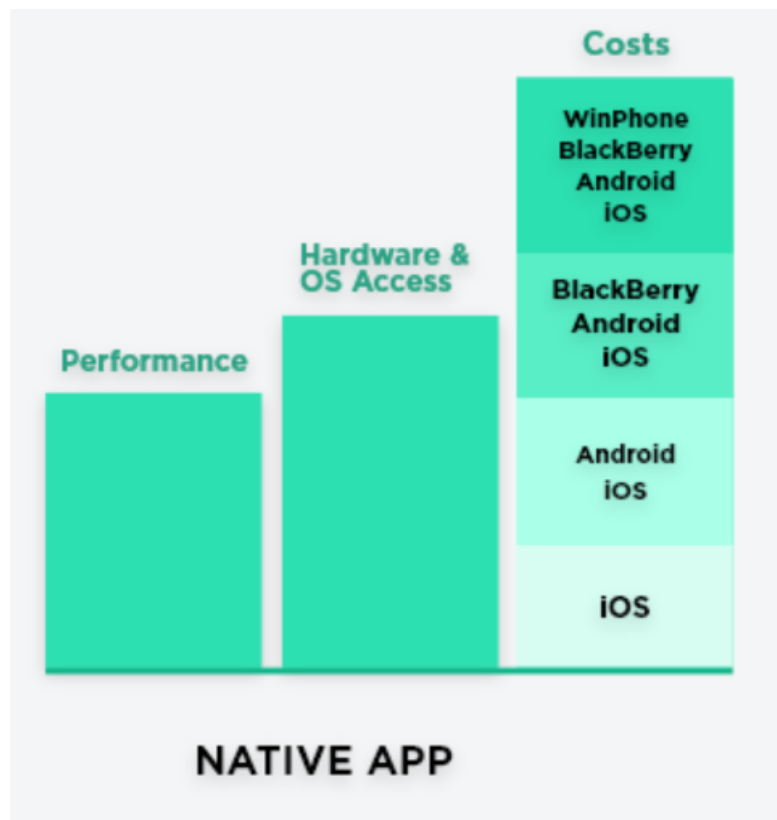
However, keep in mind that, while it is possible, it's a big risk and the success stories are usually more exceptions than the norm.

EXPENSIVE DEVELOPMENT :

The programming used in native apps is quite tricky, and developers who code in this language are scarce.

Thus, native app development requires more labor, which adds to the time and cost of development.

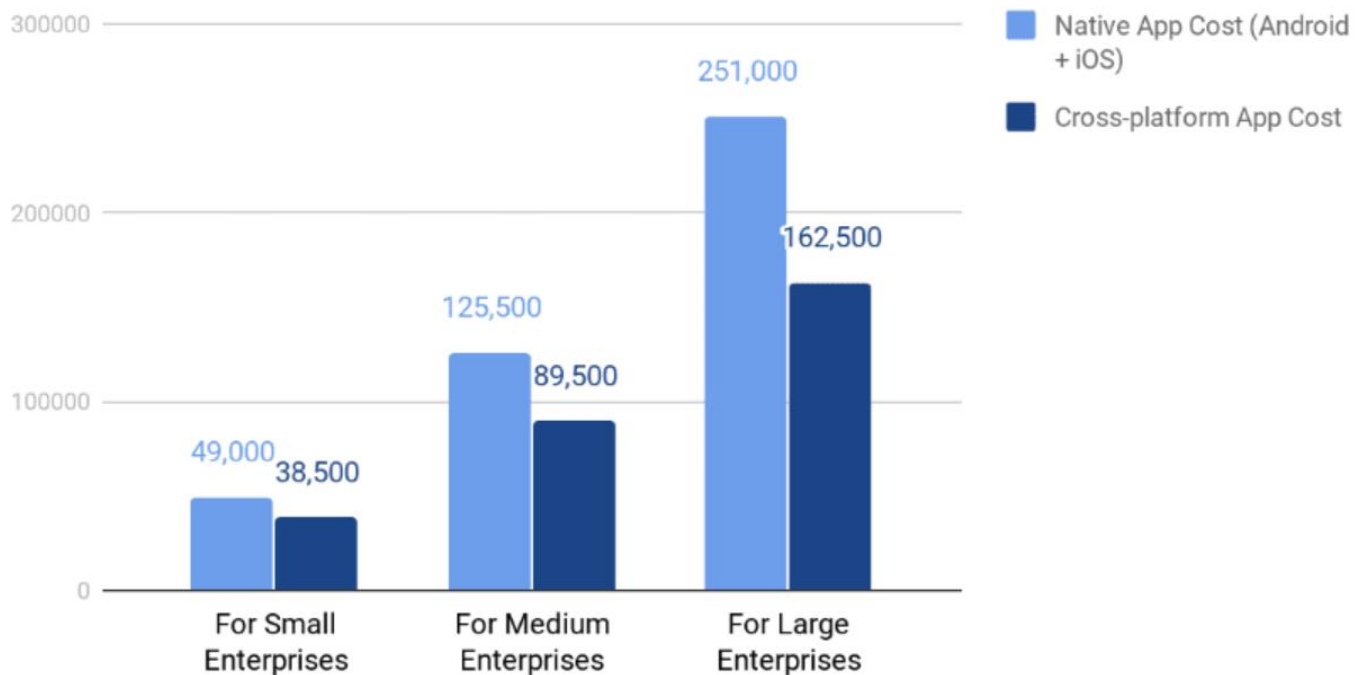
Here's a quick estimate on expensive development with Graphical representation:



When every new OS added dramatically increases the costs as well as we can see it in the graph.

Here's another statistic that confirms this with help of another app development :

Comparison Of Native vs Cross-Platform App Development Costs (USD)



The most obvious reason for this is that you need to build multiple code bases for each OS you want to develop for.

That means hiring separate teams with different expertise.

Each team will then go through the same steps the other is doing—there's rarely a way to save time and costs by combining them.

From a management standpoint, this is a big waste.

TIME CONSUMING DEVELOPMENT :

Native app development is time-consuming as the work done for a platform cannot be duplicated for another. Instead, a separate team is needed to work on the other version.

A basic native application for a single system requires at least 4 to 6 months to develop. A lot of work goes into the process, including research, planning, technical development, testing, and review.

Every platform, such as iOS and Android, needs a separate set of codes, which means more time is required, as it is equivalent to coding for two different app.

It takes about 18 weeks to develop a good quality native app. The time increases based on the project's complexity.

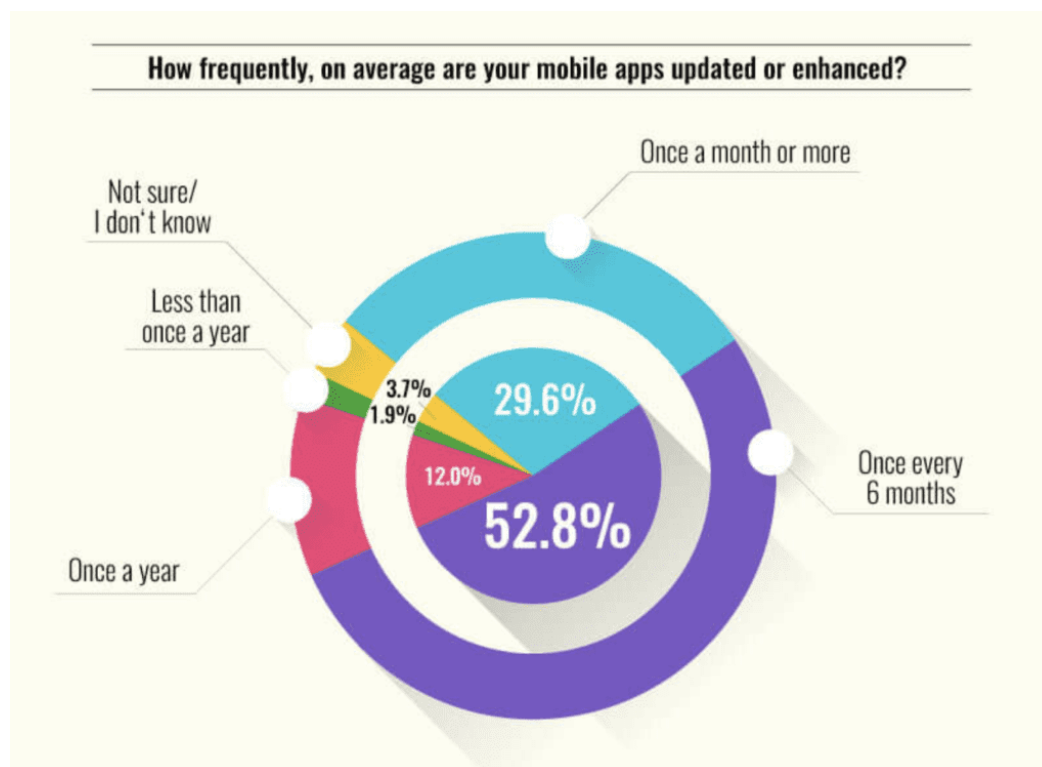
if you want to publish apps on both the App Store and Google Play, you'll need a specialized development team to build and test apps for both platforms. It may take up to 1 year to complete both versions of a native app for the two systems.

MORE SUPPORT AND MAINTENANCE :

Another big drawback of native apps is that they require an equally expensive maintenance phase. They can result in your maintenance costs accumulating.

Many developers forget that app costs don't stop once the app launches. In fact, much of the work happens during the maintenance phase.

Take a look at this survey:



An overwhelming percentage of developers (more than 50%) conduct maintenance of their apps twice a year. And almost a third of them release updates as often as monthly.

On average, yearly maintenance costs can be anywhere from 20% to 50% of the app's development budget. That's around 4,00,000 – 9,00,000 per month, according to Comentum.

The problem is that the cost of developing multiple native apps is already very high, and maintaining them can be equally astronomical.

Based on our stats from the previous section, if a native app for a large enterprise costs 200,00,000, you're looking at a yearly maintenance price tag of 40,00,000 to 10,00,000.

This also slows down bug fixes considerably.

If you spot a critical and urgent update, it will take time for you to update multiple native apps. In the meantime, your users are stuck with a subpar app that can affect their UX or even introduce dangerous vulnerabilities.

Cross-platform apps don't tend to have this problem, because of their universal code base. Changes made to it are automatically propagated to every OS version with minimal tweaking.

NATIVE APPS REQUIRE FREQUENT UPGRADES :

If a bug is fixed in native apps or a new update is about to be launched, developers have to first submit the updates to the app store and then hope that users will update their apps to the new versions.

If users don't update their app - maybe they didn't notice the update, or they don't have enough storage space - those users might abandon the app due to unfixed glitches.

Developers constantly have to worry about losing their customer base.

NATIVE APPS SHINE FOR MANY COMPANIES,NOT ALL :

Despite the fact that native apps are fast, smooth, blend in with the device's features perfectly, and can work offline, there are still many challenges. High development costs and a more time-consuming development process makes native app development a poor fit for some business owners.

Those who have smaller budgets and tighter time constraints may want to consider other options.

Companies debating whether to build a native app or web app should carefully weigh their options in order to make the best decision for their business.

CONCLUSION :

When analysing NATIVE application development, it has benefits and drawbacks, and it serves better in specific business circumstances. Native apps, being the most expensive development, offer the most reliable quality, user interaction, and performance. But the main part to understand here is not to try to pick the most affordable app development, instead, choose the one that fits the market requirements and provides the most value to the target audience.

Native application lessens the risk of the app being outlawed from the app stores. Hence, you can go with the native app if you want a larger outcome. Several businesses use these apps because of easy support and competent performance. However, aiming for wider functionality and better performance with a limited budget. Also, native app development requires more effort and expenses, but a consistent user interface and faultless performance can be worth every penny. It has many drawbacks also but it's also the most used app development in day to day life.



Source:google

THANK YOU.

Source :

<https://mdevelopers.com/blog/what-is-a-native-mobile-app-development->

<https://clutch.co/app-developers/resources/pros-cons-native-apps#:~:text=Expensive%20Development,-The%20programming%20used&text=Thus%2C%20native%20app%20development%20requires,apps%20is%20also%20very%20high>

<https://decode.agency/article/native-app-development-pros-cons/>

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